

Disproof of conjecture 00000000810: no metric star minimises the first Kirchhoff eigenvalue

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1 The conjecture and the readings refuted

Statement (English, verbatim). Definition: Kirchhoff eigenvalues of quantum graphs are the spectra of metric graphs with Kirchhoff conditions. Conjecture: Among graphs of fixed total length and vertex count, the minimum of the first eigenvalue is attained by the star graph; and the extremizer is strictly unique for more than three vertices. (The Chinese text states the same.)

Reading. The conjecture is a conjunction; we refute its first conjunct (“the minimum is attained by the star graph”) for every vertex count $n \geq 4$ and every total length $L > 0$, which refutes the conjunction. For $n \leq 3$ the star $K_{1,n-1}$ is itself a path, so nothing is claimed there. “First eigenvalue” is read as the smallest *positive* Kirchhoff eigenvalue λ_1 (the eigenvalue 0, with constant eigenfunctions, is shared by every graph). “The star graph” is read as the metric star $K_{1,n-1}$ with *any* positive edge lengths summing to L (the equilateral star included). We refute two versions:

- (R1) the competitors are all metric graphs with n vertices and total length L ;
- (R2) the competitors are only the stars $K_{1,n-1}$ of total length L . This covers any reading that excludes vertices of degree two (the path used in (R1) has such vertices).

Our competitor in (R1) is the path P_n , a simple connected tree, so (R1) is refuted for every competitor class that contains it (connected graphs, trees, simple graphs, or graphs with loops and multiple edges). *Not formalized:* the reading in which “first eigenvalue” means the bottom eigenvalue 0. Under it the minimum value 0 is attained by every graph, so uniqueness fails; this needs the non-negativity of the Kirchhoff Laplacian, which we do not formalize.

How the formal statement implies the refutation. If $\lambda_1(S) \leq \lambda_1(G)$ for every competitor G , then $\lambda_1(S) \leq \lambda$ for every positive eigenvalue λ of every competitor, because $\lambda_1(G)$ is at most each positive eigenvalue of G . The Lean theorems refute this consequence, so they refute the conjecture. No discreteness of the spectrum is used.

2 Definitions (as formalized)

Definition 1 (metric graph). A metric graph on the vertex set $\{0, \dots, n-1\}$ (**MetricGraph** n) consists of finitely many edges e , each with a tail vertex, a head vertex and a length $\ell_e > 0$. Edge e is the interval $[0, \ell_e]$; its endpoint 0 is glued to the tail and ℓ_e to the head. Loops and multiple edges are allowed. The total length is $\sum_e \ell_e$ (**totalLength**).

Definition 2 (Kirchhoff eigenvalue). $\lambda \in \mathbb{R}$ is a Kirchhoff eigenvalue (**IsEigenvalue**) if there are functions f_e with derivatives f'_e such that: (i) f_e has derivative $f'_e(x)$ and f'_e has derivative $-\lambda f_e(x)$ at every $x \in [0, \ell_e]$, within $[0, \ell_e]$ (one-sided at the endpoints), i.e. $-f''_e = \lambda f_e$; (ii) continuity: there are vertex values $F(v)$ with $f_e(0) = F(\text{tail } e)$ and $f_e(\ell_e) = F(\text{head } e)$; (iii) Kirchhoff: at every vertex v , $\sum_{\text{tail } e=v} f'_e(0) - \sum_{\text{head } e=v} f'_e(\ell_e) = 0$ (the sum of the outgoing derivatives vanishes; a loop contributes at both ends; at a degree-one vertex this is the Neumann condition); (iv) $f_e(x) \neq 0$ for some edge e and some $x \in [0, \ell_e]$. **posEig G** is the set of positive Kirchhoff eigenvalues.

Definition 3 (star and path). **star k 1**: vertices $0, \dots, k$; edge $e \in \{0, \dots, k-1\}$ joins the centre 0 to the leaf $e+1$ and has length ℓ_e . **path k L**: vertices $0, \dots, k$; edge e joins e to $e+1$ and has length L/k . Relabelling vertices and edges, or reversing an edge ($x \mapsto \ell_e - x$, which changes the sign of f'_e and swaps its two ends in (iii)), maps eigenfunctions to eigenfunctions, so every metric star with k edges has the same Kirchhoff eigenvalues as some **star k 1**.

3 Proof

Throughout, $k = n - 1 \geq 3$ is the number of edges.

Lemma 4 (**ode_cos**). Let $s, l > 0$. If $f'' = -s^2 f$ on $[0, l]$ (one-sided derivatives at the ends) and $f'(l) = 0$, then $f(x) = f(l) \cos(s(l-x))$ and $f'(x) = f(l) s \sin(s(l-x))$ on $[0, l]$.

Proof. $g = f - f(l) \cos(s(l-\cdot))$ satisfies $g'' = -s^2 g$ and $g(l) = g'(l) = 0$. The energy $E = g'^2 + s^2 g^2$ has derivative $2g'(g'' + s^2 g) = 0$, so it is constant on $[0, l]$ (Mathlib **constant_of_has_deriv_right_zero**), hence $E \equiv E(l) = 0$, so $g = g' = 0$. $\square$

Lemma 5 (**tan_add_identity**, **tan_sum_le**, **tan_sum_lt**). For $x, y \geq 0$ with $x+y < \pi/2$: $\tan x + \tan y + \tan x \tan y \tan(x+y) = \tan(x+y)$. Hence, for non-negative angles with sum $< \pi/2$, $\sum \tan \theta_i \leq \tan \sum \theta_i$; if the angles are positive and there are at least two of them, the inequality is strict.

Proof. The identity is $\tan(x+y)(1 - \tan x \tan y) = \tan x + \tan y$ after clearing the positive cosines. The inequalities follow by induction on the number of angles, since the product term is ≥ 0 (> 0 for positive angles). $\square$

Lemma 6 (**star_trig**). Let $k \geq 3$, $\theta_1, \dots, \theta_k > 0$ with $\sum \theta_e \leq \pi$, and $a_e, c \in \mathbb{R}$ with $a_e \cos \theta_e = c$ for all e and $\sum_e a_e \sin \theta_e = 0$. Then all $a_e = 0$.

Proof. Let θ_0 be a largest angle and $R = \sum_{e \neq 0} \theta_e \leq \pi - \theta_0$, a sum of at least two positive terms. For $e \neq 0$, $\theta_e < R \leq \pi - \theta_0 \leq \pi - \theta_e$, so $\theta_e < \pi/2$, $\cos \theta_e > 0$ and $a_e \sin \theta_e = c \tan \theta_e$. Also $\theta_0 < \pi$, so $\sin \theta_0 > 0$. With $T = \sum_{e \neq 0} \tan \theta_e > 0$ the Kirchhoff equation becomes $a_0 \sin \theta_0 + cT = 0$, and $c = a_0 \cos \theta_0$ gives $a_0(\sin \theta_0 + T \cos \theta_0) = 0$. We show $P = \sin \theta_0 + T \cos \theta_0 > 0$. If $\cos \theta_0 \geq 0$ this is clear. Otherwise $\theta_0 > \pi/2$, and $0 < R \leq \pi - \theta_0 < \pi/2$, so by Lemma 5 and the monotonicity of $\tan$, $T < \tan R \leq \tan(\pi - \theta_0) = -\sin \theta_0 / \cos \theta_0$. Multiplying by $\cos \theta_0 < 0$ gives $T \cos \theta_0 > -\sin \theta_0$. So $a_0 = 0$, then $c = 0$, then $a_e = 0$ for $e \neq 0$ since $\cos \theta_e > 0$. $\square$

Theorem 7 (**star_no_eigenvalue_1e**). Let $k \geq 3$ and $\ell_e > 0$ with $L = \sum \ell_e$. The star has no Kirchhoff eigenvalue λ with $0 < \lambda \leq \pi^2/L^2$.

Proof. Let $s = \sqrt{\lambda}$, so $0 < sL \leq \pi$. At the leaf $e+1$ only edge e arrives (as its head), so Kirchhoff gives $f'_e(\ell_e) = 0$, and Lemma 4 gives $f_e = a_e \cos(s(\ell_e - x))$ with $a_e = f_e(\ell_e)$. Continuity at the centre gives $a_e \cos(s\ell_e) = F(0)$; Kirchhoff at the centre gives $\sum_e f'_e(0) = s \sum_e a_e \sin(s\ell_e) = 0$. Lemma 6 with $\theta_e = s\ell_e$ gives $a_e = 0$ for all e , so $f \equiv 0$, contradicting (iv). $\square$

Theorem 8 (`path_eigenvalue`). *The path with k edges of length L/k has total length L and the eigenvalue π^2/L^2 .*

Proof. Take $f_e(x) = \cos(\frac{\pi}{L}(eL/k + x))$, i.e. $\cos(\pi t/L)$ in the arclength t , with $F(v) = \cos(\pi v/k)$. The equation holds on each edge. At a vertex v the outgoing derivative of the edge leaving v is $G(v) = -\frac{\pi}{L} \sin(\pi v/k)$, and the derivative of the edge arriving at v is $G(v)$ too, so the Kirchhoff sum is $G(v) - G(v) = 0$ at interior vertices; at the ends only one term is present and $G(0) = G(k) = 0$. Finally $f_0(0) = 1 \neq 0$. $\square$

Theorem 9 (`twoLong_eigenvalue`, `star_beaten`). *For $k = j + 2 \geq 3$ and lengths $a, a, \varepsilon, \dots, \varepsilon$ ($a, \varepsilon > 0$), the star has the eigenvalue $(\pi/2a)^2$. Consequently, if $\mu > \pi^2/L^2$, some star with k edges and total length L has a positive eigenvalue $< \mu$.*

Proof. With $s = \pi/(2a)$, take $f = \pm \cos(s(a - x))$ on the two long edges and $f = 0$ elsewhere. Then $f = 0$ at the centre (since $\cos(sa) = 0$), Neumann holds at the leaves, and the outgoing derivatives at the centre are s and $-s$. For the second claim let $r = \pi/\sqrt{\mu} < L$, $t = (L + r)/2$, $a = t/2$ and $\varepsilon = (L - t)/j$: then $2a + j\varepsilon = L$ and $(\pi/t)^2 < (\pi/r)^2 = \mu$. $\square$

Corollary 10 (`star_not_minimal`, `not_star_minimizer`, `not_star_minimizer_among_stars`). *Let $k \geq 3$ and $L > 0$. For every star S with k edges and total length L , and every $\mu \in \text{posEig } S$: $\pi^2/L^2 < \mu$; the path with $k + 1$ vertices and length L has the eigenvalue π^2/L^2 ; and another star with k edges and length L has a positive eigenvalue $< \mu$. Hence there is no star S (any lengths summing to L) whose least positive eigenvalue μ (`IsLeast (posEig S) mu`) satisfies $\mu \leq \lambda$ for all positive eigenvalues λ of all metric graphs with $k + 1$ vertices and length L (`R1`), nor for all positive eigenvalues of all such stars (`R2`).*

Remark 11. For context: Friedlander (Ann. Inst. Fourier 55 (2005) 199–211, <https://doi.org/10.5802/aif.2095>) states in the abstract: “We establish the sharp lower bound for eigenvalues of a metric graph.” Our proof does not use that paper; all steps above are proved in Lean.

4 Lean formalization and axiom audit

File `lean/Conjecture810/Basic.lean` (430 lines, namespace `C810`, only import `Mathlib`; Lean 4.33.1, `Mathlib` v4.33.1). Main theorems:

```
theorem not_star_minimizer (k : N) (hk : 3 <= k) (L : R) (hL : 0 < L) :
  not exists (l : Fin k -> R) (hl : forall e, 0 < l e), sum e, l e = L /\
  exists mu, IsLeast (posEig (star k l hl)) mu /\
  forall G : MetricGraph (k + 1), G.totalLength = L ->
    forall lam in posEig G, mu <= lam

theorem not_star_minimizer_among_stars (k : N) (hk : 3 <= k) (L : R)
  (hL : 0 < L) :
  not exists (l : Fin k -> R) (hl : forall e, 0 < l e), sum e, l e = L /\
  exists mu, IsLeast (posEig (star k l hl)) mu /\
  forall (l' : Fin k -> R) (hl' : forall e, 0 < l' e),
    sum e, l' e = L ->
      forall nu in posEig (star k l' hl'), mu <= nu
```

Supporting theorems: `star_no_eigenvalue_le`, `path_eigenvalue`, `twoLong_eigenvalue`, `star_beaten`, `star_not_minimal`, `star_trig`, `ode_cos`, `tan_sum_lt`. #print axioms for `not_star_minimizer`, `not_star_minimizer_among_stars`, `star_not_minimal`, `star_no_eigenvalue_le` and `path_eigenvalue`: [propext, Classical.choice, Quot.sound]. There is no sorry, no native_decide and no added axiom.